

- **NPU 4:** Offering three times more TOPS over its predecessor, the NPU 4 accelerates AI computations with impressively low power, reaching up to 48 TOPS.
- **IPU:** Enhances camera experiences while reducing power consumption.

Platform Controller Tile (PCT)

The PCT integrates security and connectivity, featuring upgraded components like Wi-Fi 7.0, Bluetooth 5.4, PCIe Gen5, PCIe Gen4, and Thunderbolt 4 ports. This integration ensures a robust security foundation and enhanced connectivity.

Memory on Package

This innovation enables fast data access and reduces latency, decreasing overall system power consumption, which is crucial for maintaining efficiency in high-performance applications.

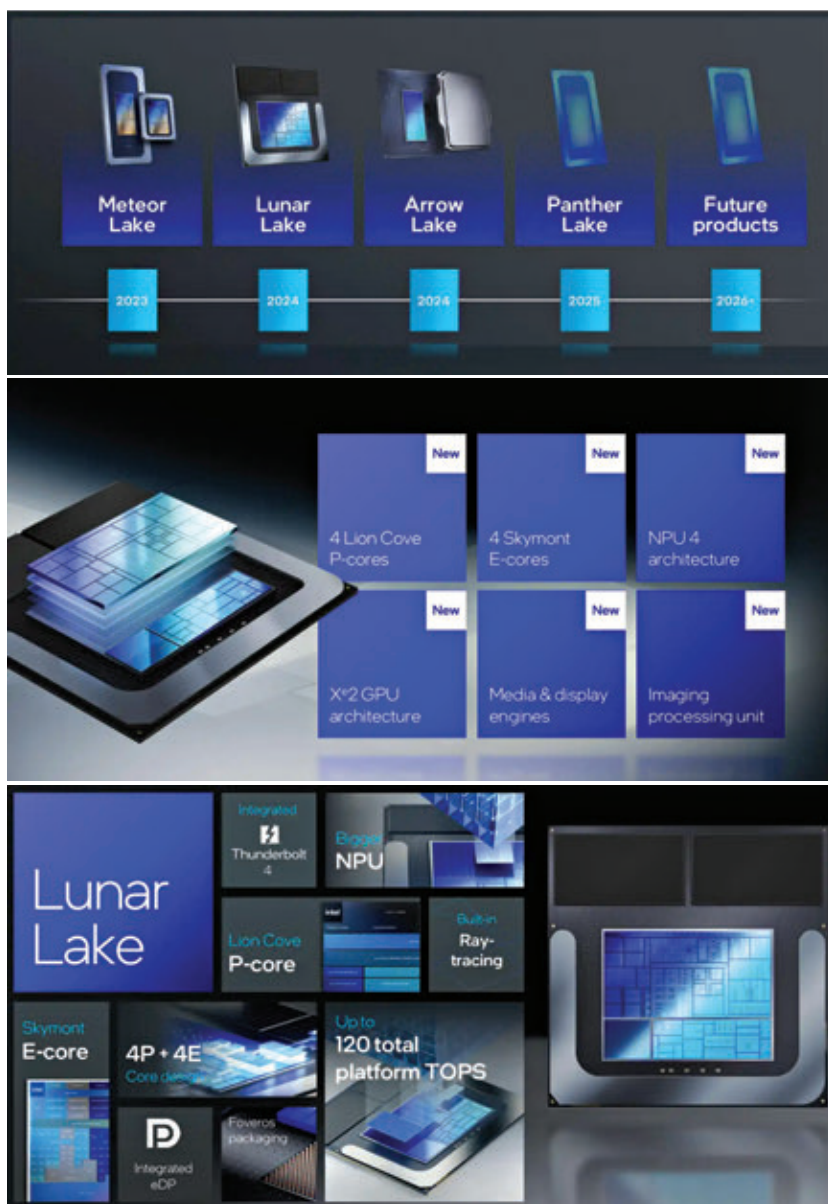
Breakthrough Efficiency and Battery Life

According to Intel, Lunar Lake sets a new standard in SoC efficiency, with a power delivery integrated controller, enhanced Thread Director, memory-side cache, and improved E-core cluster contributing to up to 60% better battery life in real-life usage scenarios. These advancements ensure that Lunar Lake-based laptops can last longer on a single charge, making them ideal for both work and play.

AI Industry Inflection Point

According to Michelle Johnston Holthaus, Executive Vice President and General Manager of Intel's Client Computing Group, the AI industry is at a massive inflection point. Holthaus believes that Lunar Lake, building on the foundation laid by Meteor Lake, will conceptually transform AI PCs and their capabilities for end-user devices.

Intel claims Lunar Lake achieves a hybrid 8-core design that offers breakthrough performance-per-watt, consuming up to 40% less power overall. The Xe2 GPU delivers powerful graphic capabilities, including real-time ray tracing and AI upscaling, with 50% more



graphics performance than its predecessor. The six NPUs on Lunar Lake (compared to just two on Meteor Lake) provide unprecedented AI compute throughput, enabling dense vector and matrix math at record-low power levels. Overall, Lunar Lake's power efficiency is nearly twice as good as previous Intel chip generations, thanks to a new PMIC power delivery system and enhanced Intel Thread Director.

Market Impact

Intel's Lunar Lake is set to power more than 80 AI PC designs, including laptops,

mini PCs, and desktop PCs, from 20 original equipment manufacturers (OEMs). These new devices are expected to ship in the third quarter of 2024, just in time for the holiday buying season. Intel aims to deploy over 40 million Core Ultra processors this year, reinforcing its commitment to bringing AI PCs to the masses swiftly and effectively. **d**

